Preparation of Liquid Dishwashing Detergent

Cheryl Cheong Kah Voon
Chua Jia Lin
Lim Yu Han
Neo Li Xin
Sabrina Heng Wei Qi

Objective

- To prepare a liquid dishwashing detergent using surfactants
- To understand the function of each chemical used
- To observe the effent of adding secondary surfactants on foam and mildness

Chemical Used

Chemical	Function
SLES	Foaming agent and mild clenser
LABS	Main grease-removing surfactant
Betaine	Foam stabilizer, mild to skin
Berol	Enhance oil and grease removal
Water	Solvent

Formulation of detergent

Chemical	Percentage (%)
SLES	4
LABS	1.5
Betaine	4
Berol	2
Water	88.5

SLES (4%) – Primary Surfactant

Role: Main cleanser & foaming agent

Why 4%:

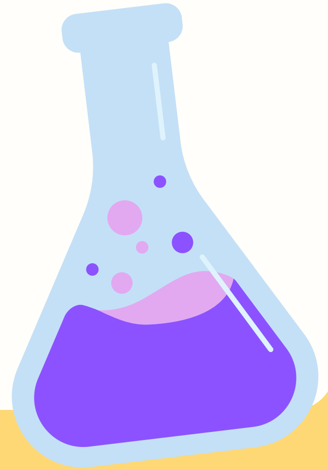
- Provides strong grease-cutting and detergency needed for dishwashing.
- Produces rich, stable foam, which consumers associate with cleaning power.
- At ~4%, it gives effective cleaning without being overly harsh on hands.
- Higher levels could cause skin dryness and unnecessary cost.

LABS (1.5%) – Secondary anionic surfactant

Role: Boosts cleaning power, especially on heavy grease

Why 1.5%:

- LABS is very strong at removing oils and fats, stronger than SLES.
- Used at a lower percentage because it is more irritating to skin and less soluble by its own
- Complements SLES by enhancing degreasing efficiency without compromising mildness.



Betaine (4%) – Amphoteric surfactant

Role: Mildness, foam booster, skin protection

Why 4%:

- Reduces irritation caused by SLES and LABS (important for hand dishwashing).
- Enhances foam volume and foam stability, even in greasy conditions.
- Improves viscosity and overall formulation stability.
- At 4%, it effectively balances performance + user comfort.

Berol (2%) – Non-ionic surfactant

Role: Grease solubilizer & performance stabilizer

Why 2%:

- Excellent at emulsifying oils that anionic surfactants may struggle with.
- Works well in hard water and across temperature variations.
- Helps prevent grease redeposition on dishes.
- Lower % is sufficient because non-ionics are highly efficient and costlier.

Thank you